
PRINCIPLES AND PRACTICES OF ACCOUNTING



*An Introduction for Readers
Who Know Nothing Yet*

A TEXTBOOK FOR FIRST-YEAR STUDENTS

BIJOY?!

TO THE READER

This book teaches accounting from the beginning. You need no earlier knowledge of business. You need only arithmetic and basic algebra. Every new word is explained in plain English when it first appears.

Bold type marks a new term at its first appearance. Bold is never used for any other purpose. **Italic type** is used for emphasis, for titles and for words from other languages.

The **narrow column** beside the text carries notes. Some notes are short summaries of the paragraph beside them, set in italic. Others give a new term in bold with a one-line meaning. A note marked **Go slowly here** tells you that the passage is new and deserves care. A note marked **Word in use** warns you that an everyday word has a special meaning in accounting.

Small raised numbers in the text refer to notes. Because this edition is set electronically, the notes are collected at the end of each chapter.

Each chapter opens with a puzzle, builds an idea in layers (the idea, an example, the rule, the exceptions), and closes with a summary, a problem, a list of common mistakes, a true story from history, and review questions. Short answers to the review questions are at the back.

Work every example with a pencil. Accounting is learned by doing. When a calculation appears, follow the same order each time: what is given, what is asked, the formula, the steps, the answer, and a check.

Partner. A person who is an owner of a partnership. (*Ch. IX*)

Partnership. A business owned by two or more people who have agreed to run it together and to share its profits. (*Ch. IX*)

Partnership deed. The written agreement setting out each partner's rights and duties. (*Ch. IX*)

Period cost. A cost of running the business in a period, such as rent or advertising; it is expensed at once. (*Ch. V*)

Plant asset. A physical asset used for many years in running the business, not held for sale, such as equipment, buildings and land. (*Ch. VIII*)

Preference share. A share that receives a fixed dividend before any dividend on ordinary shares. (*Ch. X*)

Prepaid expense. A cost paid in advance that is recorded as an asset until it is used. (*Ch. III*)

Product cost. A cost of acquiring or making goods for sale; it stays in inventory until the goods are sold. (*Ch. V*)

Receipts and payments account. A summary of the cash received and paid during a period, with no regard to the period to which items belong. (*Ch. IX*)

Residual value. The amount a business expects to recover when it disposes of a plant asset at the end of its life. (*Ch. VIII*)

Revenue. Money earned from selling goods or services. It raises equity. (*Ch. I*)

Reversing entry. An optional entry on the first day of a period that cancels an earlier accrual so that routine payments can be recorded simply. (*Ch. IV*)

Service company. A business that earns money by doing work for customers, such as a barber or a school. (*Ch. I*)

Sole trader. A business owned and run by one person, who is personally responsible for its debts. (*Ch. I*)

Special journal. A journal used for one repeated kind of transaction, such as credit sales. (*Ch. VII*)

Straight-line method. A method that charges the same depreciation every year: cost less residual value divided by useful life. (*Ch. VIII*)

T-account. A simple picture of an account with the name on top, debit on the left and credit on the right. (*Ch. II*)

Transaction. A business event stated in money that changes the assets, liabilities or equity. (*Ch. I*)

Trial balance. A list of all account balances at one date, used to test that total debits equal total credits. (*Ch. II*)

Unearned revenue. Money received before the work is done; it is a liability until the work is finished. (*Ch. III*)

Units-of-production method. A method that charges depreciation by the amount of work the asset does in each period. (*Ch. VIII*)

Useful life. The period over which a business expects to use an asset. (*Ch. VIII*)

User. A person who reads accounting information in order to make a decision. (*Ch. I*)

Weighted average. A method that gives each unit the average cost of all units available. (*Ch. VI*)

Worksheet. A large sheet on which the accountant sorts the trial balance, adjustments and statement figures before preparing the statements. (*Ch. IV*)

MATHS YOU WILL USE

Percentage. A percentage is a part of 100. To find 2 per cent of 40,000, multiply 40,000 by $2 \div 100$: the answer is 800.

Percentage change. Change divided by the starting value, times 100. A price rising from 40 to 44 changes by $4 \div 40 \times 100 = 10$ per cent.

Ratio. A ratio of 3 to 2 means five equal parts in all: three for the first share, two for the second. To split 140,000, one part is $140,000 \div 5 = 28,000$.

Average. Add the amounts and divide by their number. For weighted average, divide the total cost by the total units, not by the number of lots.

Rearranging an equation. From $A = L + E$, equity is $E = A - L$, and liabilities are $L = A - E$. Whatever is done to one side is done to the other.

Rounding. Carry full figures through a calculation and round only the final answer, to the nearest taka unless told otherwise.

Reading a graph. The horizontal axis shows the item measured, often time. The vertical axis shows the amount. A point shows the amount at that moment.

FURTHER READING

Pacioli, L. (1494). *Summa de arithmetica, geometria, proportioni et proportionalita*. Venice.

Weygandt, J., Kimmel, P. and Kieso, D. *Financial Accounting*. Wiley. Any recent edition.

IFRS Foundation. *IAS 1 Presentation of Financial Statements; IAS 2 Inventories; IAS 16 Property, Plant and Equipment*.

The Partnership Act, 1932 (Act IX of 1932), Bangladesh.

The Companies Act, 1994, Bangladesh.

A NOTE ON THE EXAMPLES

The running business in this book is **Padma Stationers**, a small shop in Dhaka owned by Nadia Rahman. It is *illustrative*: invented to show ideas at work, and not a real firm. Its figures appear as a case file: dates, amounts and decisions only.

All amounts are in **taka**, written Tk. Dates are written day, month, year. Spelling is British. Where law or standards are mentioned, the country is Bangladesh unless the text says otherwise. Laws and standards change, so confirm the current rules before relying on them.

Historical accounts in the book are true to the best of the author's knowledge. Where a date or figure could not be verified, it has been left out.

ANSWERS TO REVIEW QUESTIONS

I.1 Service (clinic), merchandising (stationery shop), manufacturing (notebook factory).

I.2 $A = L + E$. A liability is owed to outsiders by a fixed date; equity is the owner's claim, repaid last.

I.3 Tk 50,000 (80,000 – 30,000).

I.4 Common rules make reports from different businesses fair and comparable.

II.1 Asset, expense, drawings (debit); liability, capital, revenue (credit).

II.2 Dr Cash 50,000, Cr Capital 50,000; Dr Equipment 20,000, Cr Cash 20,000.

II.3 Divide by 9: $360 \div 9 = 40$ suggests a transposition. Re-add the columns first.

II.4 The trial balance lists all accounts as an internal test; the balance sheet reports assets, liabilities and equity to outsiders.

II.5 Title, debit (left) side, credit (right) side.

III.1 Accrual matches expenses to the revenue they earn. Example: wages for the last days of January are expensed in January by Dr Wages Expense, Cr Wages Payable.

III.2 Prepaid expense, unearned revenue, accrued expense, accrued revenue, and depreciation.

III.3 10 months used: Dr Insurance Expense 10,000, Cr Prepaid Insurance 10,000.

III.4 It records items that belong to a period but where cash has moved at another time, or not at all.

IV.1 A working paper used to prepare the statements; it is not formal and not part of the books.

IV.2 At period end, to test the arithmetic and sort accounts before the statements.

IV.3 Tk 593,000.

IV.4 They measure one period only; closing brings them to zero for the next period and moves profit to capital.

IV.5 Dr Capital 15,000, Cr Drawings 15,000.

V.1 They group like items under headings and subtotals so that readers can find gross profit, operating income, and current against non-current items.

V.2 Tk 180,000.

V.3 Product costs are held in inventory until sale; period costs are expensed when incurred.

V.4 Dr Purchases 8,000, Cr Equipment 8,000.

V.5 Payable is owed by the business to suppliers; receivable is owed to the business by customers.

VI.1 Goods held for sale, goods in the making, and materials awaiting use: merchandise, raw materials, work in process, finished goods.

VI.2 To serve customers at once, to buy in lots at lower prices, to guard against delay, and to keep production running.

VI.3 Too little: lost sales and stopped production. Too much: tied-up cash, storage costs, spoilage and obsolescence.

VI.4 $COGS = 100 \times 40 + 200 \times 44 = 12,800$.

VI.5 FIFO.

VII.1 A statement explaining the difference between the cash book and bank statement balances at one date.

VII.2 Deposits in transit, outstanding cheques, bank charges, interest, direct deposits and returned cheques, and errors.

VII.3 Bank notices that add to (credit memo) or subtract from (debit memo) the account.

VII.4 $33,000 + 2,000 - 5,000 = 30,000$. The statements reconcile.

VII.5 Purchases journal: 3 June, Star Pens, 12,000; Dr Purchases, Cr Accounts Payable.

VIII.1 The spreading of cost less residual value over useful life; on physical assets with a limited life used in the business.

VIII.2 To match cost with revenue, to state profit fairly, and to show book value on the balance sheet.

VIII.3 $(330,000 - 30,000) \div 10 = 30,000$. Dr Depreciation Expense 30,000, Cr Accumulated Depreciation 30,000.

VIII.4 Expense is one year's charge (income statement); accumulated depreciation is the total to date (balance sheet).

VIII.5 Rate 2 per copy; $8,000 \times 2 = 16,000$.

IX.1 Name and purpose, capitals, profit ratio, interest and salaries, drawings, admission and retirement, settlement of accounts, and ending of the firm.

IX.2 Two or more persons, an agreement, a business for profit, mutual agency and unlimited liability.

IX.3 Expiry of term, completion of venture, agreement, death or insolvency, court order, illegality.

IX.4 60,000 and 30,000.

IX.5 A summary of cash received and paid in a period, with no regard to the period to which items belong.

X.1 A legal person separate from owners, with shares and limited liability; a partnership is not separate and has unlimited liability.

X.2 Ordinary: vote, residual profit. Preference: fixed dividend first, usually no vote.

X.3 Authorised is the limit; issued is what is offered and taken; paid-up is what has been paid.

X.4 Dr Bank 20,000, Cr Share Capital 20,000.

X.5 The amount received above par value. It is part of equity, not profit.

Nadia Rahman plans to open a stationery shop in Dhaka. She has Tk 500,000 saved. A cousin asks her, a year before she opens, a plain question: "How will you know whether the shop is making money?" Nadia cannot answer. She can count the cash in the till. But cash alone does not show what she owes suppliers, what customers owe her, or what her goods are worth.

A business needs a method for answering such questions. The method has a name, **accounting**, and this chapter introduces it. Later chapters build on every idea here, so read slowly.

YOU WILL NEED: nothing earlier. This is where the book begins. Chapter II uses the equation of this chapter at once.

WHAT A BUSINESS DOES

A **business** means an organisation that sells goods or services to customers in order to earn a profit. A **profit** means the money left after all costs of the business are paid. If costs are larger than the money earned, the result is a **loss**. Nadia's shop is a business. It buys pens and notebooks and sells them for more than they cost.

Businesses are grouped by what they sell. A **service company** sells work, such as a barber, a clinic or a photocopy bureau. A **merchandising company** buys finished goods and sells them again without change, as a stationery shop does. A **manufacturing company** makes goods from materials and sells them, as a factory making notebooks does.

The three kinds are different in one important way. A service company holds no goods for sale. A merchandising company holds goods bought from others. A manufacturing company holds materials, goods half-made, and goods finished. Each extra kind of goods needs extra records, which is why this book moves from service to merchandising, and touches manufacturing in Chapter VI.

A business also has a form of ownership. A **sole trader** is owned by one person. A partnership is owned by two or more people (Chapter IX). A company is a separate legal body owned by shareholders (Chapter X). Padma Stationers begins as a sole trader, and later chapters show how its owner might change the form.

WHAT ACCOUNTING IS

Accounting means recording, sorting and reporting the money facts of a business so that people can make decisions. A shop sells forty notebooks on a Monday. Accounting records the sale, adds it to the week's sales, and reports the total to the owner. The record is of no use unless someone acts on it.

Accounting has often been called the language of business. The description is fair in three ways. Accounting turns many different events into one common unit, money. It is understood by owners, bankers and tax officers alike. And, like any language, it has rules of grammar that must be followed. The description fails in one way: a language can say anything, but accounting reports only what can be measured in money.

A business sells something to earn a profit.

Business. An organisation that sells goods or services to customers in order to earn a profit.

Businesses fall into three kinds.

Service company. A business that earns money by doing work for customers, such as a barber or a school.

Each kind has its own accounts.

Merchandising company. A business that buys finished goods and sells them again without making any change.

Manufacturing company. A business that turns materials into new goods and then sells those goods.

Three common forms of ownership.

Sole trader. A business owned and run by one person, who is personally responsible for its debts.

Accounting records, sorts and reports money facts.

Accounting. The process of recording, sorting and reporting the money facts of a business so that people can make decisions.

The idea is old. People called it the language of business.

GLOSSARY

Padma Stationers has done well, and Nadia wants to open ten shops. She needs Tk 100 million, more than she and her partner can supply. She could ask a hundred friends to invest. But who would invest if, should the business fail, their homes could be taken? The company form answers that fear.

YOU WILL NEED: equity and capital (Chapter I), and partnership and unlimited liability (Chapter IX).

WHAT A COMPANY IS

A company is a legal person.

Company. A business that the law treats as a person separate from its owners, with its capital divided into shares.

Owners risk only what they invest.

Limited liability. The rule that owners lose at most the amount they agreed to pay for their shares.

A **company** means a business that the law treats as a person separate from its owners, with its capital divided into **shares**. In Bangladesh companies are formed under the Companies Act, 1994. The company owns property, makes contracts and can be sued, in its own name. It does not die when an owner dies.

The owners are called *shareholders*. Their **limited liability** means they can lose at most the sum they agreed to pay for their shares. This differs from a partner's unlimited liability. Companies are *private* when they may not offer shares to the public, and *public* when they may. Equity in a company is called *shareholders' equity* or stockholders' equity.

KINDS OF SHARES

Ordinary and preference shares.

Ordinary share. A share that gives a vote and a share of profits after all other claims, with no fixed dividend.

Preference share. A share that receives a fixed dividend before any dividend on ordinary shares.

Capital has several names.

An **ordinary share** gives its owner a vote and a share in profits, but it receives a dividend only after others are paid. A **preference share** has a fixed dividend, paid before ordinary shares. A **dividend** means a share of profit paid to shareholders. Preference shareholders usually have no vote.

Several terms describe the capital. *Authorised capital* is the largest amount the company may issue, set in its founding documents. *Issued capital* is the part offered and taken. *Paid-up capital* is the part for which shareholders have actually paid. The **par value**, or face value, is the amount printed on a share, often Tk 10. A share sold above par is sold at a *premium*.

ISSUING SHARES

GO SLOWLY HERE.

Payment may be in stages.

GO SLOWLY HERE.

Worked example: shares issued at a premium.

Shares are often paid for in stages. The buyer pays *application money* when applying, *allotment money* when shares are given, and may pay one or more *calls* later. A separate account holds each stage until the shares are allotted.

GIVEN. Padma Ltd issues 10,000 ordinary shares of Tk 10 at Tk 12 each. Payment: Tk 3 on application, Tk 5 on allotment (including the premium of Tk 2), and Tk 4 on the first and final call. All money is received. FIND. The journal entries and the totals. STEPS. Application money: 10,000 × 3 = 30,000. Allotment: 10,000 × 5 = 50,000. Call: 10,000 × 4 = 40,000.

THE ACCOUNTING EQUATION

GO SLOWLY HERE.

Everything in accounting begins with a simple balance.

GO SLOWLY HERE.

Asset. Something of value that a business owns and can use in future, such as cash or equipment.

Liability. An amount the business owes to someone outside it.

Equity. The owner's claim on the business: what is left after liabilities are taken from assets.

GO SLOWLY HERE.

Accounting equation. Assets equal liabilities plus equity. It holds after every transaction.

GO SLOWLY HERE.

Equity changes by four items.

Revenue. Money earned from selling goods or services. It raises equity.

Expense. A cost of running the business, such as rent or wages. It lowers equity.

A transaction is a money event that changes the equation.

Transaction. A business event stated in money that changes the assets, liabilities or equity.

Given and find: the first worked example.

Each line balances.

WORD IN USE. Capital. In daily life, money. In accounting, the owner's investment in the business.

Think of a household. Your family owns a house worth Tk 1,000,000. A bank lent you Tk 400,000 of the price. The part you truly own is the rest, Tk 600,000. What you own equals what you owe plus what is yours. That is the whole idea of this section.

An **asset** means something of value that a business owns, such as cash, goods and equipment. A **liability** means an amount the business owes to others, such as a bank loan. **Equity** means the owner's claim: what remains after liabilities are subtracted from assets. Equity is also called owner's equity, or stockholders' equity in a company. Equity and liability are alike because both are claims on the assets. They differ because a liability must be repaid by a fixed date, but equity is repaid only after all debts.

The relation is the **accounting equation**. In words: assets equal liabilities plus equity. In symbols: A = L + E. Here A stands for assets, L for liabilities and E for equity. The equation holds at all times, because every asset was paid for either by borrowing or by the owner.

Equity is changed by four items. The owner's investment raises it, and **drawings**, meaning money the owner takes out for private use, lower it. **Revenue** means money earned by selling goods or services; it raises equity. An **expense** means a cost of running the business, such as rent; it lowers equity. Revenue minus expenses is the profit. We write the full form: A = L + (Capital + Revenue – Expenses – Drawings).

A **transaction** means an event, stated in money, that changes assets, liabilities or equity. Buying equipment is one. Agreeing to hire a worker next month is not, because no money value has changed yet. A transaction is also supported by a document, such as a receipt, and it affects the business itself.

GIVEN. During January 2025 Padma Stationers, then a photocopy service, has seven transactions: (1) Nadia invests Tk 500,000 cash; (2) buys equipment for Tk 150,000 cash; (3) buys supplies on credit for Tk 20,000; (4) earns fees of Tk 40,000 in cash; (5) pays rent Tk 12,000; (6) pays Tk 10,000 to the supplier; (7) draws Tk 15,000 for private use. FIND. The totals of A, L and E after each transaction. FORMULA. A = L + E. STEPS. See the table below.

PADMA STATIONERS: EQUATION AFTER EACH TRANSACTION (TK)			
Transaction	Assets	Liabilities	Equity
1 Investment	500,000	0	500,000
2 Equipment, cash	500,000	0	500,000
3 Supplies on credit	520,000	20,000	500,000
4 Fees earned	560,000	20,000	540,000
5 Rent paid	548,000	20,000	528,000
6 Supplier paid	538,000	10,000	528,000
7 Drawings	523,000	10,000	513,000

SUMMARY

A non-trading concern reports a surplus or deficit, on an accrual basis. A partnership rests on an agreement, written in a deed. Profit is shared by interest, salary and ratio. On liquidation, assets are sold, debts paid, and the rest goes to partners.

THINK IT THROUGH

Two partners share profits equally. A has capital Tk 100,000, B Tk 50,000, and profit is Tk 40,000. If there is no deed, how is it divided? (Tk 20,000 each. Without agreement, the law gives equal shares regardless of capital.)

WHERE PEOPLE GO WRONG

Students forget that unlimited liability reaches private property. Others pay out the partners before paying creditors. Debts always come first.

IN HISTORY

Before 1932 the law of partnership in this region was contained in Chapter XI of the Indian Contract Act of 1872. The Indian Partnership Act of 1932 gave it a separate statute. In Bangladesh it continues as the Partnership Act, 1932.

REVIEW QUESTIONS

1. Describe the elements of a partnership deed.
2. What are the characteristics of a partnership?
3. What are the common reasons for liquidation?
4. Net profit Tk 90,000, ratio 2 to 1. Share the profit.
5. What is a receipts and payments account?

¹ The Partnership Act, 1932 (Act IX of 1932), sections 4 and 5. The firm is not a legal person separate from its partners.

¹ Standards are revised. A reader should check the current requirements of the Financial Reporting Council and the Institute of Chartered Accountants of Bangladesh.

A cricket club collects fees, pays a groundsman and buys balls. It makes no profit, yet its members want to know whether it is spending wisely. Meanwhile Nadia has invited a friend, Imran, to join her shop. They must decide how to share what the shop earns. Each case needs accounts suited to its purpose.

YOU WILL NEED: revenue, expense and accrual (Chapters I and III), and the statements (Chapters IV and V).

THE NON-TRADING CONCERN

A club exists to serve its members.

Non-trading concern. An organisation, such as a club or charity, that provides service to members or the public and does not exist to earn profit.

Two statements are prepared.

Receipts and payments account. A summary of the cash received and paid during a period, with no regard to the period to which items belong.

GO SLOWLY HERE.

Worked example: converting cash to accrual.

A **non-trading concern** is an organisation that exists to serve its members or the public, and not to earn profit. Clubs, schools and charities are examples. They have no sales and no profit. Their summary of the year is a *surplus* if income is greater than expense, or a *deficit* if it is less.

A **receipts and payments account** summarises the cash received and paid, whatever period the items belong to. It is a cash statement. An **income and expenditure account** is the equivalent of the income statement. It uses the accrual basis: it includes only the income and expenses that belong to the year, and leaves out capital items, such as the purchase of equipment.

GIVEN. The club received subscriptions of Tk 60,000 in the year. At the start, members owed Tk 2,000 from the previous year; at the end, they owed Tk 5,000. The club paid salaries of Tk 24,000 and still owed Tk 2,000 at year end. It paid rent of Tk 18,000 and bought sports equipment of Tk 15,000. **FIND.** The surplus. **STEPS.** Subscriptions earned = 60,000 – 2,000 + 5,000 = 63,000. Salaries incurred = 24,000 + 2,000 = 26,000. Rent = 18,000. The equipment is a capital item and is left out. Surplus = 63,000 – 26,000 – 18,000 = 19,000. **CHECK.** The adjustments are the same as accrual adjustments in Chapter III.

THE PARTNERSHIP

Partnership begins with an agreement.

Partnership. A business owned by two or more people who have agreed to run it together and to share its profits.

Partner. A person who is an owner of a partnership.

Characteristics of a partnership.

A **partnership** means a business owned by two or more people who have agreed to run it together and share its profits. The people are called **partners**, and together they are the *firm*. In Bangladesh the matter is governed by the Partnership Act of 1932. It holds that a partnership arises from an agreement, and not from family status or the mere joint ownership of property.¹

A partnership has these features. There are at least two partners. There is an agreement, which may be spoken or written. The business is carried on for profit. Each partner can bind the others in the ordinary work of the business, a feature called *mutual agency*. Each partner is responsible, without limit, for the debts of the firm, which is called **unlimited liability**: creditors may look to the private property of any partner.

Nadia buys a copier for Tk 150,000 and pays cash. The records must say two things: the business now has a copier, and it now has less cash. If only one is written, the books no longer balance. Over a year a shop has thousands of such events. How can they be written so that mistakes reveal themselves?

The answer was published in Venice more than five hundred years ago. It is called double entry, and it is the subject of this chapter.

YOU WILL NEED: the accounting equation, assets, liabilities, equity, revenue, expense and transaction (Chapter I).

THE ACCOUNTING CYCLE

The cycle is a fixed list of steps.

Accounting cycle. The steps repeated every period, from recording transactions to closing the books.

Each step guards accuracy.

The **accounting cycle** means the steps an accountant repeats every period. First, analyse each transaction. Second, record it in the journal. Third, post it to the ledger. Fourth, prepare the unadjusted trial balance. Fifth, make adjusting entries. Sixth, prepare the adjusted trial balance. Seventh, prepare the financial statements. Eighth, close the temporary accounts. This chapter teaches steps one to four.

Each step checks the one before it. Analysis ensures that the event is a transaction. The journal records it in order of date. The ledger sorts it by account. The trial balance tests whether debits equal credits. Later steps update and report. Skipping any step lets errors pass unseen.

ACCOUNTS, DEBITS AND CREDITS

An **account** means a record of all changes to one kind of item, such as Cash or Rent Expense. Think of a notebook with a page headed *Cash*. Every time cash arrives, a line is added on the left. Every time cash leaves, a line is added on the right. That page is the idea of an account.

The page is drawn as a **T-account**, named for its shape. It has three parts: the *title* at the top, the *debit side* on the left, and the *credit side* on the right. **Debit** means the left side of an account; **credit** means the right side. The words do not mean good or bad. They mean left and right.

Six types of account exist. Assets, drawings and expenses are *increased* by a debit and decreased by a credit. Liabilities, equity (capital) and revenue are *increased* by a credit and decreased by a debit. The rule follows from the equation: $A = L + E$ has assets on the left, so they grow on the left.

GO SLOWLY HERE.

An account is a named record.

Account. A record of all changes to one kind of item, such as Cash or Rent Expense.

WORD IN USE. Account. In daily life, a bank record. In accounting, a named record of one kind of item.

GO SLOWLY HERE.

The T-account has two sides and a name.

T-account. A simple picture of an account with the name on top, debit on the left and credit on the right.

GO SLOWLY HERE.

The sign of increase depends on the account type.

SUMMARY

A plant asset is physical, long-lasting and used in the business. Its cost is spread over its useful life by depreciation. Straight-line, units-of-production and declining-balance are the main methods. Accumulated depreciation is the total charged to date.

THINK IT THROUGH

A van costs Tk 800,000, with residual value Tk 80,000 and life 6 years. Find the straight-line charge and book value after 2 years. ($720,000 \div 6 = 120,000$; book value $800,000 - 240,000 = 560,000$.)

WHERE PEOPLE GO WRONG

Students often subtract nothing for residual value, or charge depreciation on land. Others record accumulated depreciation as an expense. It is a deduction from the asset.

IN HISTORY

In the nineteenth century, British railway companies owned vast tracks, bridges and engines. Many paid dividends without providing for wear and replacement. When the assets decayed, the money to renew them was not there. The lesson helped bring depreciation into ordinary practice.

REVIEW QUESTIONS

- 1. What is depreciation, and on which assets is it charged?
- 2. Point out the reasons for charging depreciation.
- 3. Cost Tk 330,000, residual Tk 30,000, life 10 years. Straight-line depreciation and journal entry?
- 4. How does depreciation expense differ from accumulated depreciation?
- 5. A machine of 64,000 expected copies and cost Tk 128,000 with residual Tk 0 makes 8,000 copies. Depreciation?

THE LEDGER

The **ledger** means the complete set of accounts. Posting means copying each journal line to its account. The journal tells the story by date; the ledger gathers every change to Cash on one page. After posting, the *balance* of an account is found by subtracting the smaller side from the larger.

GIVEN. The seven journal entries above. **FIND.** The balance of Cash. **STEPS.** Debit side: $500,000 + 40,000 = 540,000$. Credit side: $150,000 + 12,000 + 10,000 + 15,000 = 187,000$. Balance: $540,000 - 187,000 = 353,000$, a debit balance. **ANSWER.** Cash is Tk 353,000. **CHECK.** Assets after transaction 7 in Chapter I were 523,000, and Cash 353,000 + Supplies 20,000 + Equipment 150,000 = 523,000.

THE TRIAL BALANCE

A **trial balance** means a list of all ledger balances at one date, in two columns, to test that total debits equal total credits. It is not a financial statement. The balance sheet reports position to outsiders and divides assets, liabilities and equity; the trial balance is an inside check that lists every account, including revenue and expenses.

TRIAL BALANCE, 31 JANUARY 2025 (TK)

Account	Debit	Credit
Cash	353,000	
Supplies	20,000	
Equipment	150,000	
Accounts Payable		10,000
Capital		500,000
Drawings	15,000	
Fees Revenue		40,000
Rent Expense	12,000	
Totals	550,000	550,000

A balanced trial balance does not prove the books are right. It misses an entry left out, an entry posted to the wrong account, an error repeated on both sides, and two errors that cancel. It does catch a one-sided posting and most errors of addition.

When the totals disagree, work in order. Re-add both columns. Find the difference. If it is divisible by 9, suspect a *transposition*, such as 1,520 written as 1,250 (difference 270, and $270 \div 9 = 30$). If the difference equals one item, suspect that item was left out; if it is twice an item, suspect that item was placed in the wrong column. Then check that every ledger balance was copied. Correct an error by a new journal entry, not by erasing.

SUMMARY

Every transaction is analysed, journalised, posted and tested. Debits equal credits. Assets, expenses and drawings rise by debit. Liabilities, capital and revenue rise by credit. The trial balance tests arithmetic but not judgement.

The ledger sorts by account.

Ledger. The complete set of accounts of a business, in which journal entries are sorted by account.

Posting the Cash account.

The trial balance tests the books.

Trial balance. A list of all account balances at one date, used to test that total debits equal total credits.

Debits equal credits: the test passes.

Limits of the test.

Finding an error.

Nadia buys a copier for Tk 440,000. It will serve the shop for about ten years. If the whole price were treated as an expense in year one, that year would show a heavy loss and the next nine a false gain. The copier helps earn revenue in every year it works. Its cost should therefore be shared among those years.

YOU WILL NEED: asset, expense, adjusting entries and accumulated depreciation (Chapters I and III).

PLANT ASSETS

Plant assets are long-lasting tools.

Plant asset. A physical asset used for many years in running the business, not held for sale, such as equipment, buildings and land.

Cost includes all costs of readiness.

A **plant asset** means a physical asset used for many years to run the business and not held for sale. Examples are a copier, shelves, a van, a building and land. A van bought by a van dealer to sell is inventory. The same van bought by the shop to deliver goods is a plant asset. Use decides the class.

The recorded cost of a plant asset is everything paid to get it ready for use: the price, taxes that cannot be recovered, freight, and installation. Later repairs that only keep it working are expenses, not part of cost. For a machine bought at Tk 600,000, with freight of Tk 20,000 and installation of Tk 30,000, the cost is Tk 650,000.

DEPRECIATION

GO SLOWLY HERE.

Depreciation spreads cost over use.

Depreciation. The spreading of the cost of a plant asset, less its residual value, over the years it is expected to be used.

GO SLOWLY HERE.

Depreciation is not a valuation, and not a saving of cash.

Depreciation means spreading the cost of a plant asset, less any amount expected to be recovered at the end, over the years in which it is used. It is charged on assets that are *physical*, that have a *limited life*, and that are *used in the business*. Land is not depreciated, because it does not wear out. Inventory is not depreciated, because it is sold.

Two misunderstandings are common. Depreciation does not measure what the asset could be sold for today. It is a way of dividing cost. And it does not set cash aside: the entry Dr Depreciation Expense, Cr Accumulated Depreciation moves no cash at all.

There are three reasons. It matches the cost of the asset to the revenue it helps to earn. It gives a fairer profit each year. And it shows the asset on the balance sheet at a sensible figure, its book value, which is cost less accumulated depreciation.

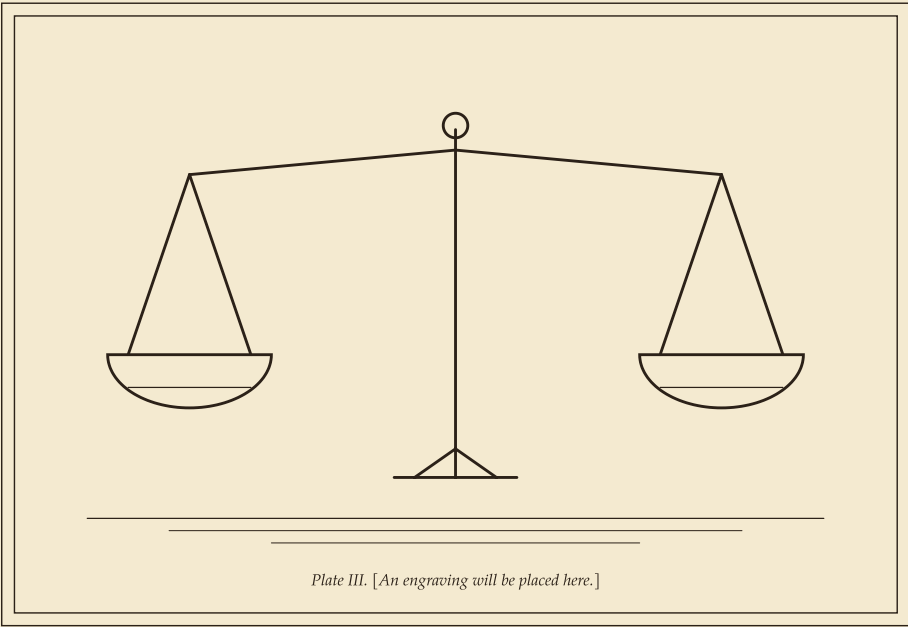
Three estimates fix the yearly charge. The first is *cost*. The second is **residual value**, the amount expected from sale or scrap at the end of life. The third is **useful life**, the period of expected use, measured in years or in units of output. The last two are judgements, so depreciation is an estimate. Wear, age and new inventions, called obsolescence, all shorten life.

Reasons for charging depreciation.

Three factors fix the charge.

Residual value. The amount a business expects to recover when it disposes of a plant asset at the end of its life.

Useful life. The period over which a business expects to use an asset.



CHAPTER III

Adjusting the Accounts

On 31 January the shop's books say one thing, and the truth is another.

This chapter explains why the books must be corrected at the end of each period, the kinds of adjustment, and how to record them, using Padma Stationers at the end of January 2025.

THINK IT THROUGH

Cash book Tk 20,000; bank statement Tk 21,500. Outstanding cheques are Tk 3,000, and the bank charged Tk 500 not yet recorded. Is there a deposit in transit? (Adjusted cash book 19,500. Bank side 21,500 – 3,000 = 18,500, so a deposit in transit of 1,000.)

WHERE PEOPLE GO WRONG

A frequent error is to make a journal entry for an outstanding cheque. It needs none. Another is to add an item to the wrong side: bank charges lower the cash book, while outstanding cheques lower the bank balance.

IN HISTORY

The Bank of England was founded in 1694 by royal charter. As cheques spread in later centuries, customers began to keep their own record of cheques written and to compare it with the bank's statement, which is the practice this chapter teaches.

REVIEW QUESTIONS

1. What is a bank reconciliation statement?
2. What are the common causes of differences?
3. What are credit and debit memos?
4. Cash book Tk 30,000, bank statement Tk 33,000; outstanding cheques Tk 5,000, deposit in transit Tk 2,000. Prove the difference.
5. Prepare a purchases journal entry: bought Tk 12,000 goods on credit from Star Pens on 3 June.

ADJUSTING THE BOOKS OF PADMA STATIONERS

UNADJUSTED TRIAL BALANCE, 31 JANUARY 2025 (TK)

Account	Debit	Credit
Cash	360,000	
Supplies	20,000	
Prepaid Insurance	6,000	
Equipment	150,000	
Accounts Payable		10,000
Unearned Fees		9,000
Capital		500,000
Drawings	15,000	
Fees Revenue		64,000
Rent Expense	12,000	
Wages Expense	20,000	
Totals	583,000	583,000

The starting point.

The six facts found at month end.

GO SLOWLY HERE.

Method: ask what the balance should be, then record the difference.

GIVEN. (a) Supplies on hand are Tk 8,000. (b) The Tk 6,000 insurance covers six months from 1 January. (c) The equipment cost Tk 150,000, will last 10 years (120 months) and be worth Tk 30,000 at the end. (d) Wages of Tk 3,000 are owed to workers for 27 to 31 January. (e) Half of the Tk 9,000 advance has now been earned. (f) Fees of Tk 6,000 are earned but not yet billed.

FIND. The six adjusting entries. METHOD. For each account ask two questions: what is the balance now, and what should it be? The entry is the difference. STEPS. (a) Supplies used = 20,000 – 8,000 = 12,000. (b) Insurance used = 6,000 ÷ 6 = 1,000. (c) Depreciation = (150,000 – 30,000) ÷ 120 = 1,000. (d) Given. (e) 9,000 ÷ 2 = 4,500. (f) Given.

ADJUSTING ENTRIES, 31 JANUARY 2025 (TK)

Particulars	Debit	Credit
(a) Supplies Expense	12,000	
Supplies		12,000
(b) Insurance Expense	1,000	
Prepaid Insurance		1,000
(c) Depreciation Expense	1,000	
Accumulated Depreciation		1,000
(d) Wages Expense	3,000	
Wages Payable		3,000
(e) Unearned Fees	4,500	
Fees Revenue		4,500
(f) Accounts Receivable	6,000	
Fees Revenue		6,000

Accumulated depreciation is a contra account.

Accumulated depreciation. The total depreciation charged on an asset so far. It is subtracted from the asset's cost.

Accumulated depreciation means the total depreciation charged so far. It is a *contra account*: it has a credit balance and is subtracted from the asset it belongs to. The asset itself stays at its cost. The balance sheet shows the cost, the accumulated depreciation, and the difference, called the book value.

Padma Stationers grows. It makes dozens of credit purchases and sales a week. If every one is written in the general journal, with two lines each, the book fills fast and the same words are copied again and again. A sensible accountant sorts the events by kind and gives each kind its own book.

YOU WILL NEED: the journal, ledger, accounts payable and receivable (Chapter II), and the periodic system (Chapter V).

SPECIAL JOURNALS

A **special journal** is a journal used for one repeated kind of transaction. The *sales journal* records credit sales of goods. The *purchases journal* records credit purchases of goods. The *cash receipts* and *cash payments journals* record cash events. Anything else goes to the *general journal*. The main gain is time: only the total of a column is posted, once a month.

The buyers on credit are many, so a **subsidiary ledger** keeps one account for each customer, and another for each supplier. The general ledger holds a single **control account**, Accounts Receivable or Accounts Payable, whose balance must equal the total of the subsidiary accounts. Each posting is made to both, and the agreement is a check.

PURCHASES JOURNAL, PADMA STATIONERS, MAY 2025 (TK)

Date	Supplier	Invoice	Amount
3 May	Dhaka Paper Co	101	40,000
8 May	Star Pens	102	25,000
Total (Dr Purchases, Cr Accounts Payable)			65,000

SALES JOURNAL, PADMA STATIONERS, MAY 2025 (TK)

Date	Customer	Invoice	Amount
4 May	Alam Stores	201	30,000
14 May	Rahim Traders	202	18,000
Total (Dr Accounts Receivable, Cr Sales)			48,000

THE CASH BOOK

The **cash book** is a book that records all cash received and paid, and also serves as the Cash account itself, so it is posted to no separate Cash account. It has a debit side for receipts and a credit side for payments. A discount allowed is shown on the debit side, and a discount received on the credit side.

CASH PAYMENTS JOURNAL, PADMA STATIONERS, MAY 2025 (TK)

Date	Paid to	Payables Dr	Other Dr	Discount Cr	Cash Cr
12 May	Dhaka Paper Co	40,000		800	39,200
18 May	Rent expense		8,000		8,000
Totals		40,000	8,000	800	47,200

Each journal handles one kind of event.

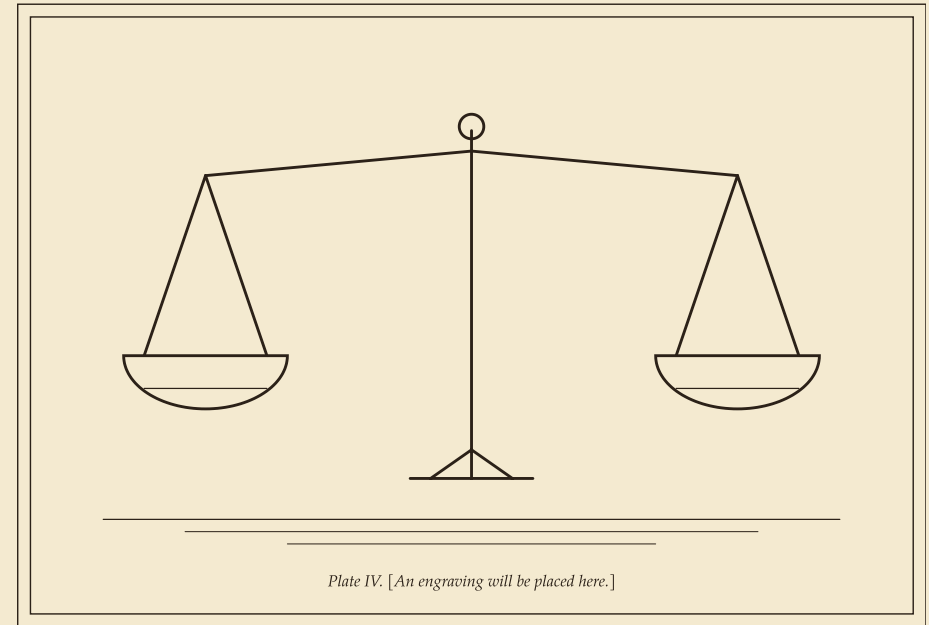
Special journal. A journal used for one repeated kind of transaction, such as credit sales.

Subsidiary ledgers and control accounts.

Control account. A general ledger account whose balance equals the total of many individual customer or supplier accounts.

The cash book combines journal and ledger.

Cash book. A book that records all cash received and paid and also serves as the Cash account.



CHAPTER IV

The Worksheet and the Closing of the Books

How can an accountant see the whole year on one sheet before writing a single statement?

This chapter explains the ten-column worksheet, the financial statements drawn from it, and the closing and reversing entries that prepare the books for the next period.

IN HISTORY

The Revenue Act of 1939 in the United States is generally given as the law that allowed LIFO for tax purposes to businesses in general. Since firms had to use it in their books if they used it for tax, accounting for stock became a matter of tax planning as well as measurement.

REVIEW QUESTIONS

- 1. Define and classify inventory.
- 2. Why do organisations maintain inventory?
- 3. What are the consequences of too little and too much inventory?
- 4. Using the data in this chapter, find FIFO COGS if 300 units were sold.
- 5. Which method gives the highest profit when prices rise?

PADMA STATIONERS: TEN-COLUMN WORKSHEET, 31 JANUARY 2025 (TK)

Account	Unadj Dr	Unadj Cr	Adj Dr	Adj Cr	Adj TB Dr	Adj TB Cr	Income Dr	Income Cr	Bal Sheet Dr	Bal Sheet Cr
Cash	360,000				360,000				360,000	
Accounts Receivable			(f) 6,000		6,000				6,000	
Supplies	20,000			(a) 12,000	8,000				8,000	
Prepaid Insurance	6,000			(b) 1,000	5,000				5,000	
Equipment	150,000				150,000				150,000	
Accumulated Depreciation				(c) 1,000		1,000				1,000
Accounts Payable		10,000				10,000				10,000
Wages Payable				(d) 3,000		3,000				3,000
Unearned Fees		9,000	(e) 4,500			4,500				4,500
Capital		500,000				500,000				500,000
Drawings	15,000				15,000				15,000	
Fees Revenue		64,000		(e,f) 10,500		74,500		74,500		
Rent Expense	12,000				12,000		12,000			
Wages Expense	20,000		(d) 3,000		23,000		23,000			
Supplies Expense			(a) 12,000		12,000		12,000			
Insurance Expense			(b) 1,000		1,000		1,000			
Depreciation Expense			(c) 1,000		1,000		1,000			
Totals	583,000	583,000	27,500	27,500	593,000	593,000	49,000	74,500	544,000	518,500
Net income							25,500			25,500
Totals							74,500	74,500	544,000	544,000

How to read the sheet and check it.

Read a row from left to right. Take Supplies: the trial balance shows a debit of 20,000; adjustment (a) credits 12,000; the adjusted debit is 8,000; it is an asset, so it goes to the balance sheet debit column. The first check is that each pair totals equally. The last check is that net income is the same number in the income statement and balance sheet columns. Here both are Tk 25,500.

Net income balances two pairs.

The income statement columns show credits of 74,500 and debits of 49,000. The difference of 25,500 is net income. It is added to the debit column to make the pair equal. In the balance sheet columns, the same figure is added to the credit side, because net income increases capital.

THE STATEMENTS

The statements come straight from the sheet.

Income statement. A report of revenue, expenses and net income for a period of time.

From the income columns.

The income statement reports revenue, expenses and net income for a period. The balance sheet reports the position at one date. They are linked: net income from the first is added to equity in the second. This is how they are connected.

INCOME STATEMENT, MONTH ENDED 31 JANUARY 2025 (TK)

Item	Tk
Fees revenue	74,500
Less expenses:	
Wages	23,000
Rent	12,000
Supplies used	12,000
Insurance	1,000
Depreciation	1,000
Total expenses	49,000
Net income	25,500

Padma Stationers starts the quarter with 100 notebooks that cost Tk 40 each. It then buys 200 more at Tk 44, and later 100 at Tk 48. Over the quarter, 250 notebooks are sold. The notebooks are identical, and no one can say which were sold. Yet the cost of the 250 must be stated, and the cost of the 150 left on the shelf. The answer depends on a rule.

YOU WILL NEED: cost of goods sold, gross profit and the periodic system (Chapter V).

WHAT INVENTORY IS

Inventory is a current asset.

Inventory (types). Goods held for sale, goods being made, and materials waiting to be used.

Work in process and finished goods differ.

Inventory is a current asset, because it is expected to become cash within a year. Its type depends on the business. A merchandising company holds *merchandise inventory*, goods bought for resale. A manufacturing company holds three kinds: *raw materials* (cloth, paper), *work in process* (goods partly made), and *finished goods* (goods complete and unsold). Supplies used in the office are not for sale and are kept apart.

The difference between the last two kinds is one of completion. A notebook factory's paper and card are raw materials. Sheets cut and printed but not yet bound are work in process. Bound notebooks in the warehouse are finished goods. Each kind carries its own accumulated cost.

WHY INVENTORY IS HELD, AND THE RISK OF HOLDING IT

Reasons for holding stock.

Too little and too much both cost money.

Organisations hold inventory for four reasons. Customers must be served at once, or they go elsewhere. Buying in larger lots often lowers the price per unit. Stock protects against delays by suppliers. And a manufacturer needs materials ready so production does not stop.

Too little inventory leads to lost sales, unhappy customers and stopped work. Too much ties up cash, needs storage, and risks theft, spoilage and goods that go out of fashion. Management therefore aims at the amount that serves customers without waste.

VALUING INVENTORY

GO SLOWLY HERE.

Cost flows need a rule.

Cost flow assumption. A rule stating which unit costs are treated as sold first, used to divide the cost of goods between sold and unsold.

GO SLOWLY HERE.

FIFO sells the oldest costs first.

FIFO. First in, first out: the oldest costs are treated as sold first, so inventory carries the newest costs.

Weighted average. A method that gives each unit the average cost of all units available.

When unit costs differ, the cost of goods available must be divided between goods sold and goods unsold. A **cost flow assumption** is a rule for doing so. The rule need not match the physical movement of goods. Four methods are in common use. *Specific identification* tracks the real cost of each unit, and suits cars or jewels. The three that follow suit identical goods.

FIFO means first in, first out: the oldest costs are treated as sold first. **LIFO** means last in, first out: the newest costs are treated as sold first. **Weighted average** gives every unit the same average cost, found by dividing total cost by total units.

IN HISTORY

Francesco Datini, a merchant of Prato in Italy, died in 1410. His business archive survives in Prato, with some five hundred account books and about 150,000 letters. It shows a merchant who balanced his books at set dates and compared branch with branch.

REVIEW QUESTIONS

1. What is a worksheet? Is it a formal document?
2. Why and when is it prepared?
3. In the worksheet above, what is the sum of the adjusted trial balance debits?
4. Why are revenue and expense accounts closed?
5. Write the closing entry for drawings of Tk 15,000.

SUMMARY

Cost of goods sold = opening inventory + net purchases – closing inventory. A classified income statement shows gross profit, then operating income, then net income. A classified balance sheet separates current from non-current items. Product costs wait in inventory; period costs are expensed at once.

THINK IT THROUGH

Opening inventory is Tk 40,000, purchases 300,000, purchase returns 10,000, freight-in 5,000, closing inventory 50,000. Find COGS. (40,000 + 300,000 – 10,000 + 5,000 – 50,000 = 285,000.)

WHERE PEOPLE GO WRONG

Students often subtract purchase returns from sales. They belong in the cost of goods. Others place freight-in with selling expenses, though it is a cost of getting the goods.

IN HISTORY

In the early 1770s, during a trade depression, the potter Josiah Wedgwood studied his costs and found that he had been underestimating them. He used what he learned to adjust output and prices. The episode is often cited as an early use of cost analysis in manufacturing.

REVIEW QUESTIONS

1. What are a classified income statement and a classified balance sheet, and what do they serve?
2. Compute gross profit: net sales Tk 500,000, COGS Tk 320,000.
3. Differentiate product cost from period cost.
4. A purchase of goods for Tk 8,000 was debited to Equipment. Give the correction.
5. What is the difference between accounts payable and accounts receivable?

Padma Stationers now buys notebooks and pens from wholesalers and sells them to students. At the end of the year Nadia asks two questions. How much did the goods she sold cost her? And how much did she earn above that cost? A service business has no such questions. A merchandising business must answer them first.

YOU WILL NEED: revenue, expense, the income statement and the balance sheet (Chapters I and IV). The figures below are a fresh illustrative set for the year 2025.

BUYING AND SELLING GOODS

Inventory means goods held for sale. When goods are bought they are an asset. When they are sold, their cost becomes an expense called **cost of goods sold** (COGS). **Gross profit** means sales minus cost of goods sold. It shows what the goods earned before the cost of running the shop.

Four accounts record the buying. *Purchases* holds the cost of goods bought. *Purchase returns* holds goods sent back to the supplier, and lowers cost. *Freight-in* holds the cost of carrying the goods to the shop, and raises cost. *Sales returns* holds goods customers send back, and lowers sales. Under the *periodic system* used here, goods on hand are counted at the end of the period.

FORMULA. In words: cost of goods sold equals opening inventory, plus net purchases, minus closing inventory. Net purchases are purchases, less returns, plus freight-in. In symbols: $\text{COGS} = \text{OI} + (\text{P} - \text{R} + \text{F}) - \text{CI}$. The opening inventory is what was on hand at the start, and the closing inventory is what is counted at the end.

THE CLASSIFIED INCOME STATEMENT

A **classified statement** groups items of the same kind under headings, with subtotals. It serves readers: a banker can find the gross profit at a glance, and an owner can compare selling costs with administrative costs. A list of twenty items in no order tells the reader little. The classified income statement has these parts: net sales, cost of goods sold, gross profit, operating expenses (selling and administrative), operating income, other income and expenses, and net income.

GO SLOWLY HERE.

Goods bought are an asset until sold.

Inventory. Goods a business holds for sale in the ordinary course of trade.

Cost of goods sold. The cost of the goods that were sold during the period.

GO SLOWLY HERE.

Four accounts record the buying.

WORD IN USE. Goods. In daily life, any things. In accounting, items bought for resale.

GO SLOWLY HERE.

The cost of goods sold is found by a short sum.

Classification groups like items.

Classified statement. A statement in which items of the same kind are grouped under headings and subtotals.